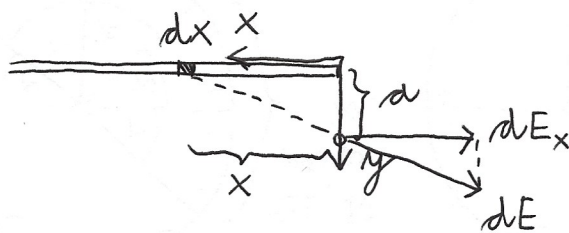
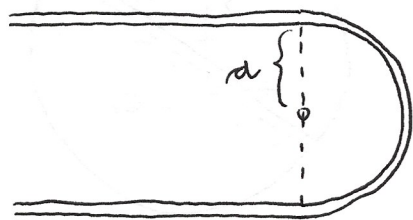


1. KOLOKVIJ

1 NABITA NITKA V OBLIKU ČRKE "U"

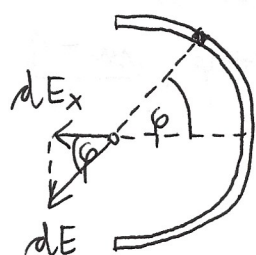


$$+ \left\{ dE = \frac{\mu dx}{4\pi\epsilon_0 (x^2 + a^2)} \right.$$

$$+ \left\{ dE_x = \frac{x}{\sqrt{x^2 + a^2}} dE = \frac{\mu}{4\pi\epsilon_0} \frac{x dx}{(x^2 + a^2)^{3/2}} \right.$$

$$+ \left\{ E_x = \frac{\mu}{4\pi\epsilon_0} \int_0^\infty \frac{x dx}{(x^2 + a^2)^{3/2}} = \boxed{\frac{\mu}{4\pi\epsilon_0 a}} \right.$$

$$\frac{1}{2} \int_0^\infty \frac{d(x^2 + a^2)}{(x^2 + a^2)^{3/2}} = \frac{1}{2} (-2) \frac{1}{\sqrt{x^2 + a^2}} \Big|_0^\infty = \frac{1}{a}$$



$$+ \left\{ dE = \frac{\mu a d\phi}{4\pi\epsilon_0 a^2} \right.$$

$$+ \left\{ dE_x = dE \cos\phi = \frac{\mu}{4\pi\epsilon_0 a} \cos\phi d\phi \right.$$

$$+ \left\{ E_x = \frac{\mu}{4\pi\epsilon_0 a} \int_{-\pi/2}^{\pi/2} \cos\phi d\phi = \boxed{\frac{\mu}{2\pi\epsilon_0 a}} \right.$$

$$\sin\phi \Big|_{-\pi/2}^{\pi/2} = 2$$

SKUPAJ:

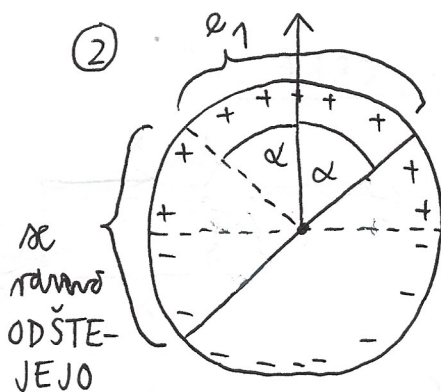
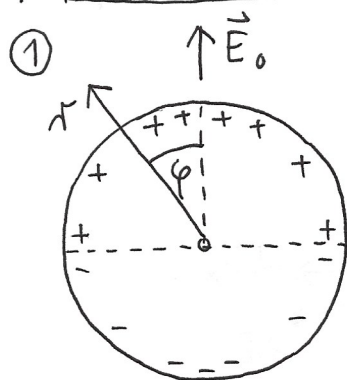
$$+ \left\{ \begin{array}{l} - \text{prispevki v smeri y se zaradi simetrije odštejejo: } \boxed{E_{y,y} = 0} \end{array} \right.$$

$$+ \left\{ \begin{array}{l} - \text{v smeri x sta dva prispevka enakega vodnika in} \\ \text{en prispevek polkrožnega vodnika:} \end{array} \right.$$

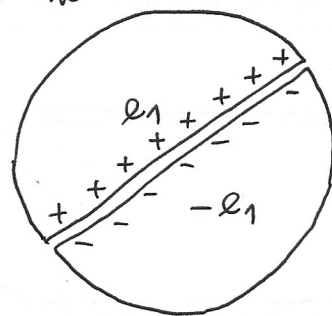
$$E_{0,x} = -2 \cdot \frac{\mu}{4\pi\epsilon_0 a} + \frac{\mu}{2\pi\epsilon_0 a} = 0, \quad \boxed{E_{0,x} = 0}$$

karže v smeri -x

2. PREPOLOVLJENI VALJ



③ PLOŠČATI kondenzator



① $\nabla^2 V = 0$ zunaj valja

1/4 {

RP1: $V(a, \varphi) = 0$

RP2: $V(r \rightarrow \infty, \varphi) = -E_0 r \cos \varphi \Rightarrow$ v rešitvi sta le člena s $\cos \varphi$, $m=1$

$V(r, \varphi) = (A_1 r + B_1 r^{-1}) \cos \varphi$

+ { RP2 $\Rightarrow A_1 = -E_0$

+ { RP1 $\Rightarrow -E_0 a + \frac{B_1}{a} = 0 \Rightarrow B_1 = E_0 a^2$

$$V(r, \varphi) = -E_0 r \cos \varphi + \frac{E_0 a^2}{r} \cos \varphi = E_0 \left(\frac{a^2}{r} - r \right) \cos \varphi$$

+ { $\sigma = \epsilon_0 \left(-\frac{\partial V}{\partial r} \right)_{r=a} = \epsilon_0 E_0 \cos \varphi \left(\frac{a^2}{r^2} + 1 \right)_{r=a} = 2\epsilon_0 E_0 \cos \varphi$

② + { del nabojev v smeri 2. $\left(\frac{\pi}{2} - \alpha \right)$ se ODŠTEJE,

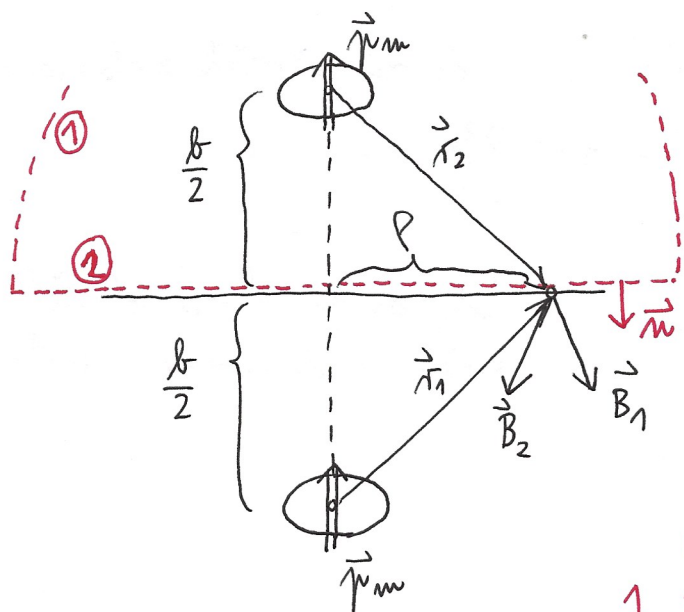
+ { $q_1 = \int \sigma dS = \int_{-\alpha}^{\alpha} 2\epsilon_0 E_0 \cos \varphi \cdot la d\varphi = 2\epsilon_0 E_0 la \sin \varphi \Big|_{-\alpha}^{\alpha} = 4\epsilon_0 E_0 la \sin \alpha$

③ izračunajmo površino PLOŠČATI KONDENZATORA

+ { $E_1 = \frac{q_1}{\epsilon_0 S} = \frac{4\epsilon_0 E_0 la}{\epsilon_0 \cdot 2al} \sin \alpha = 2E_0 \sin \alpha$

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3. MAGNETNA SILA MED TOKOVNIMA ŽANKAMA



$$\vec{B}_{DIP} = \frac{\mu_0}{4\pi} \frac{3\vec{r}(\vec{p}_{mm} \cdot \vec{r}) - \vec{p}_{mm} r^2}{r^5}$$

$$\vec{p}_{mm} = \begin{bmatrix} \theta \\ p_{mm} \end{bmatrix} \rightarrow \text{v ravnini} \rightarrow \text{v smeri osi z}$$

$$\vec{r}_{1,2} = \begin{bmatrix} p \\ \pm \frac{b}{2} \end{bmatrix}, \quad r_{1,2}^2 = p^2 + \frac{b^2}{4}$$

$\frac{1}{4}$

$$3\vec{r}_{1,2}(\vec{p}_{mm} \cdot \vec{r}_{1,2}) - \vec{p}_{mm} r_{1,2}^2 =$$

$$= \pm \frac{3}{2} p_{mm} b \begin{bmatrix} p \\ \pm \frac{b}{2} \end{bmatrix} - \left(p^2 + \frac{b^2}{4}\right) \begin{bmatrix} \theta \\ p_{mm} \end{bmatrix} =$$

$$= \begin{bmatrix} \pm \frac{3}{2} p_{mm} b p \\ \frac{1}{2} p_{mm} b^2 - p_{mm} p^2 \end{bmatrix} \rightarrow \text{se odštejeta} \rightarrow \text{se seštejeta}$$

$$\vec{B} = \vec{B}_{DIP,1} + \vec{B}_{DIP,2} = \frac{\mu_0}{4\pi} \frac{2 \left(\frac{1}{2} b^2 - p^2\right) p_{mm}}{\left(p^2 + \frac{b^2}{4}\right)^{\frac{5}{2}}} \hat{e}_z$$

$$\vec{F}_{mm} = \frac{1}{\mu_0} \oint \left[\vec{B} (\vec{B} \cdot \vec{n}) - \frac{1}{2} B^2 \vec{n} \right] dS, \quad \left[\text{PLOSKEV: simetrijska (vodoravna) ravnina, sklenjena v neskončnosti} \right]$$

$$\vec{F}_{mm,1} = \theta, \text{ saj polje močno pada}$$

$$\textcircled{2} \quad \vec{B} \parallel \vec{n} \Rightarrow \vec{F}_{m2} = \frac{1}{2\mu_0} \vec{n} \int B^2 dS, \quad \vec{n} = -\hat{e}_z$$

$$+ \left\{ F_{m2} = \frac{1}{2\mu_0} \left(\frac{\mu_0}{2\pi} \right)^2 p_{mm}^2 \int_0^\infty \frac{\left(\frac{1}{2} b^2 - p^2\right)^2}{\left(p^2 + \frac{b^2}{4}\right)^5} 2\pi p dp = \frac{\mu_0 p_{mm}^2}{4\pi \cdot 2} \int_{\frac{b^2}{4}}^\infty \frac{\left(\frac{3}{4} b^2 - u\right)^2}{u^5} du =$$

$$\frac{1}{4} \left\{ = \frac{\mu_0 p_{mm}^2}{8\pi} \left[\frac{9}{16} b^4 \left(-\frac{1}{4}\right) u^{-4} - \frac{3}{2} b^2 \left(-\frac{1}{3}\right) u^{-3} + \left(-\frac{1}{2}\right) u^{-2} \right]_{\frac{b^2}{4}}^\infty = \right.$$

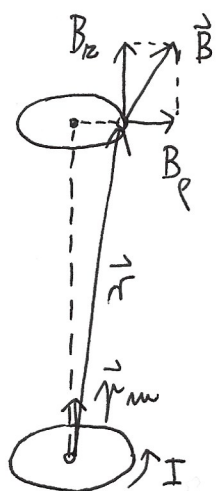
$$= \frac{\mu_0 p_{mm}^2}{8\pi} \left[\frac{9}{64} b^4 \frac{64 \cdot 4}{b^8} - \frac{1}{2} b^2 \frac{64}{b^6} + \frac{1}{2} \frac{16}{b^4} \right] = \frac{3\mu_0 p_{mm}^2}{2\pi b^4}$$

$$+ \left\{ \vec{F}_{mm} = - \frac{3\mu_0 (\pi a^2 I)^2}{2\pi b^4} \hat{e}_z = - \frac{3\pi\mu_0 I^2 a^4}{2 b^4} \hat{e}_z, \text{ sila je PRIVLAČNA}$$

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3) ALTERNATIVNI REŠITVI

1) DIREKTNO RAČUNANJE SILE



$$\vec{r} = \begin{bmatrix} \rho \\ z \end{bmatrix} \quad \text{valjne koordinate}$$

$$\vec{r} = \begin{bmatrix} a \\ b \end{bmatrix}, \quad \vec{p}_m = \begin{bmatrix} \theta \\ p_m \end{bmatrix}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{3 \begin{bmatrix} a \\ b \end{bmatrix} p_m b - \begin{bmatrix} \theta \\ p_m \end{bmatrix} (a^2 + b^2)}{(a^2 + b^2)^{\frac{5}{2}}} =$$

$$= \frac{\mu_0}{4\pi} p_m \frac{\begin{bmatrix} 3ab \\ 2b^2 - a^2 \end{bmatrix}}{(a^2 + b^2)^{\frac{5}{2}}}$$

$$p_m = \pi a^2 I \Rightarrow B_\phi = \frac{\mu_0}{4\pi} p_m \frac{3ab}{b^5} = \frac{3\mu_0 \pi}{4\pi} \frac{I a^3}{b^4}$$

$$F_m = I \cdot 2\pi a \cdot B_\phi = \boxed{\frac{3\pi\mu_0}{2} \frac{I^2 a^4}{b^4}}, \quad \text{enak rezultat!} \quad \boxed{1}$$

2) PREKO MAGNETNE ENERGIJE

$$W_m = -p_m B = -\frac{\mu_0}{2\pi} \frac{p_m^2}{b^3}$$

$$\downarrow$$

$$\frac{\mu_0}{4\pi} \frac{2p_m^2}{b^3}$$

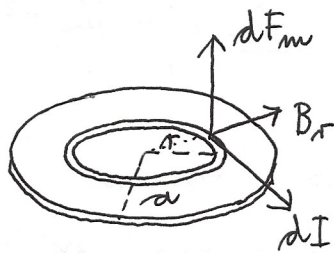
$$dW_m = F_m db \Rightarrow F_m = \frac{dW_m}{db} = \frac{3\mu_0}{2\pi} \frac{p_m^2}{b^4} = \boxed{\frac{3\mu_0 \pi}{2} \frac{I^2 a^4}{b^4}}$$

enak rezultat!

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Tudi ti dve rešitvi sta prinesli POLNO ŠTEVILO TOČK!
(Čeprav sta seveda bistveno lažji.)

4 KOVANEC V GRADIENTU MAGNETNEGA POLJA



$$\left\{ \begin{aligned} \Phi &= \pi r^2 B_z(z) & \text{skozi obroček polmera } r \\ \dot{\Phi} &= \pi r^2 \dot{B}_z = \pi r^2 \frac{\partial B_z}{\partial z} \cdot v = U_i \end{aligned} \right.$$

$$dI = U_i d\left(\frac{1}{R}\right)$$

tok, ki ga skozi obroček požene inducirana napetost

$$\left\{ d\left(\frac{1}{R}\right) = \frac{\oint dS}{l} = \frac{\oint h dr}{2\pi r} = \frac{h}{2\pi} \frac{dr}{r} \right.$$

$$\left\{ dI = \frac{1}{2} \frac{h}{\pi} v \frac{\partial B_z}{\partial z} r dr \right.$$

$$dF_m = 2\pi r (B_r) dI, \text{ iščemo } B_r$$

$$B_r(r) = B_r(\theta) + r \frac{\partial B_r}{\partial r} = r \frac{\partial B_r}{\partial r} \downarrow$$

$$\vec{\nabla} \cdot \vec{B} = \frac{1}{r} \frac{\partial (r B_r)}{\partial r} + \frac{\partial B_z}{\partial z} = \frac{1}{r} B_r + \frac{\partial B_r}{\partial r} + \frac{\partial B_z}{\partial z} = \frac{2}{r} B_r + \frac{\partial B_z}{\partial z} = 0$$

$$\left\{ \begin{aligned} B_r &= -\frac{r}{2} \frac{\partial B_z}{\partial z} \end{aligned} \right.$$

$$\rightarrow dF_m = -\frac{\pi}{2} h v \left(\frac{\partial B_z}{\partial z}\right)^2 r^3 dr$$

$$F_m = -\frac{\pi}{2} h v \left(\frac{\partial B_z}{\partial z}\right)^2 \int_0^a r^3 dr = -\frac{\pi}{8} h v \left(\frac{\partial B_z}{\partial z}\right)^2 a^4$$

$$\left\{ \text{po drugi strani je } F_m = m v = \rho \pi a^2 h \underbrace{\frac{\partial v}{\partial z}}_{v'} \cdot \underbrace{\frac{\partial z}{\partial t}}_v, \text{ se pravi} \right.$$

$$\left\{ -\frac{\pi}{8} h v a^4 B_z'^2 = \rho \pi a^2 h v' v \right.$$

$$\left\{ v' = -\frac{v a^2}{8\rho} B_z'^2 \Rightarrow \text{hitrost je odvisna le od koordinate } z! \right.$$

(ne od časa)

$$\left\{ \Delta v = -\frac{v a^2}{8\rho} \underbrace{B_z'^2}_{\left(\frac{B_0}{D}\right)^2} \underbrace{\Delta z}_D = -\frac{v a^2 B_0^2}{8\rho D} \Rightarrow \right.$$

$$\boxed{v_0 = \frac{v a^2 B_0^2}{8\rho D}}$$

$$\boxed{v_0 \sim 1 \frac{m}{s}}$$

1